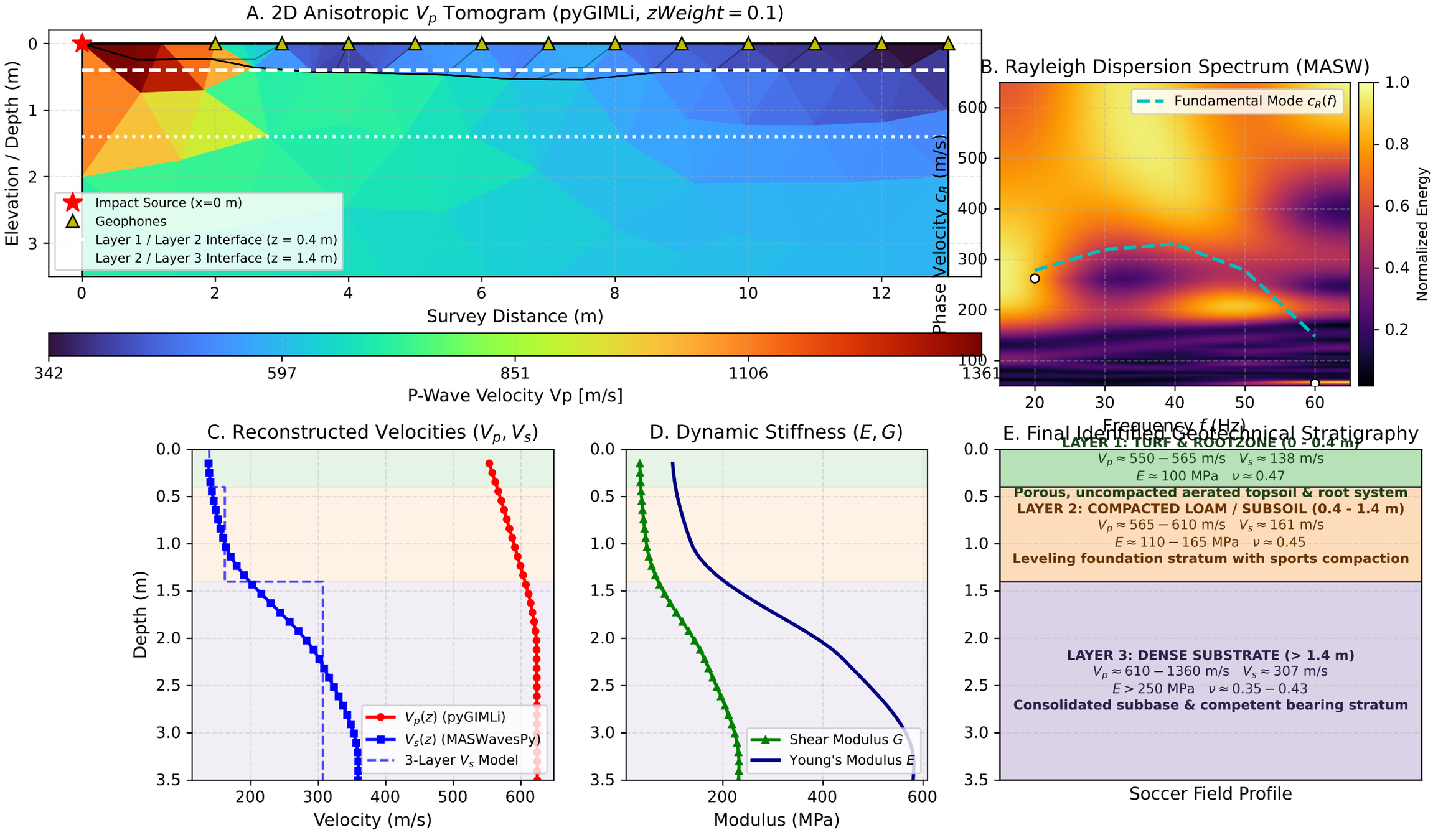

A. 2D Anisotropic V_p Tomogram (pyGIMLi, $zWeight = 0.1$)
Elevation / Depth (m)
Survey Distance (m)
Impact Source (x=0 m)
Geophones
Layer 1 / Layer 2 Interface (z = 0.4 m)
Layer 2 / Layer 3 Interface (z = 1.4 m)
P-Wave Velocity Vp [m/s]
B. Rayleigh Dispersion Spectrum (MASW)
Fundamental Mode $c_R(f)$
Phase Velocity c_R (m/s)
Frequency f (Hz)
Normalized Energy
C. Reconstructed Velocities (V_p, V_s)
$V_p(z)$ (pyGIMLi)
$V_s(z)$ (MASWavesPy)
3-Layer V_s Model
Depth (m)
Velocity (m/s)
D. Dynamic Stiffness (E, G)
Shear Modulus G
Young's Modulus E
Modulus (MPa)
E. Final Identified Geotechnical Stratigraphy
LAYER 1: TURF & ROOTZONE (0 - 0.4 m)
$V_p \approx 550 - 565$ m/s $V_s \approx 138$ m/s
$E \approx 100$ MPa $\nu \approx 0.47$
Porous, uncompacted aerated topsoil & root system
LAYER 2: COMPACTED LOAM / SUBSOIL (0.4 - 1.4 m)
$V_p \approx 565 - 610$ m/s $V_s \approx 161$ m/s
$E \approx 110 - 165$ MPa $\nu \approx 0.45$
Leveling foundation stratum with sports compaction
LAYER 3: DENSE SUBSTRATE (> 1.4 m)
$V_p \approx 610 - 1360$ m/s $V_s \approx 307$ m/s
$E > 250$ MPa $\nu \approx 0.35 - 0.43$
Consolidated subbase & competent bearing stratum
Soccer Field Profile